LOCAL FINITENESS OF THE SUPPORT OF A COHERENT MODULE AND FINITENESS OF THE LENGTH AT A GENERIC POINT

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file chow.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. Preamble provenance: the pinned chapter file opens with \input{preamble} and carries no preamble text of its own, so the project's single Stacks-wide preamble staged at sources/stacks-chow/preamble.tex is excerpted verbatim as three spans, lines 8-12 (the verbatim package together with the project's own reference, slogan and history display-environment definitions, including the author's commented-out older form of the reference environment), lines 14-18 (Xy-pic setup) and lines 154-217 (the native theorem declarations and mathematical macros). The lines 8-12 span is required because a counted statement or its surrounding text may use those display environments, which the project typesets as comments. Nothing was added, removed or redefined; the wrapper class and the amsmath/amssymb/amsthm/graphicx base packages are supplied by the assembler.

Editorial note. Same-chapter pointers are located at their \label lines inside the pinned chapter file; cross-chapter pointers are located in the project's own tag list staged in the snapshot (sources/stacks-chow/tags.txt), at the verbatim lines carrying their canonical tags.

Allowed standard prerequisites (editorial).

- The support of a coherent module is closed, and its reduced structure is a reduced closed subscheme. Cohomology of Schemes, label coherent-lemma-coherent-support-closed (proof line 1723); Schemes, label schemes-lemma-reduced-closed-subscheme (proof line 1726). Realises the support as a reduced closed subscheme, so that statements about locally finite collections of components and about multiplicities apply to it.
- Local finiteness of the irreducible components of a closed subscheme, finiteness of multiplicities, specialisation of points and finiteness of length over a ring whose module is supported at the closed point. Divisors, label divisors-lemma-components-locally-finite (proof line 1728); Same chapter chow.tex, label lemma-multiplicity-finite (proof line 1730); Schemes, label schemes-lemma-specialize-points (proof line 1737); Algebra, label algebra-lemma-support-point (proof line 1740). The first two give parts (1) and (3) directly, and the last two show that the stalk at the generic point is a finite module over a local ring supported only at the closed point, hence of finite length.

Source reference. Original lemma 00L5: sources/stacks-chow/tags.txt, source lines 775–775. Navigation pointer for \ref{algebra-lemma-support-point} used inside the counted span (line 1740); the argument that consumes it is retained in the span.

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Source reference. Original lemma 01Y5: sources/stacks-chow/tags.txt, source lines 2544–2544. Navigation pointer for \ref{coherent-lemma-coherent-support-closed} used inside the counted span (line 1723); the argument that consumes it is retained in the span.

Source reference. Original lemma 0BE1: sources/stacks-chow/tags.txt, source lines 14334–14334. Navigation pointer for \ref{divisors-lemma-components-locally-finite} used inside the counted span (line 1728); the argument that consumes it is retained in the span.

Source reference. Original lemma 02QT: sources/stacks-chow/pinned-original-source.tex, source lines 1629–1629. Navigation pointer for \ref{lemma-multiplicity-finite} used inside the counted span (line 1730); the argument that consumes it is retained in the span.

Source reference. Original lemma 01J3: sources/stacks-chow/tags.txt, source lines 2000–2000. Navigation pointer for \ref{schemes-lemma-reduced-closed-subscheme} used inside the counted span (line 1726); the argument that consumes it is retained in the span.

Source reference. Original lemma 01J7: sources/stacks-chow/tags.txt, source lines 2004–2004. Navigation pointer for \ref{schemes-lemma-specialize-points} used inside the counted span (line 1737); the argument that consumes it is retained in the span.

Source reference. Original reference 02QL: sources/stacks-chow/pinned-original-source.tex, source lines 1372–1372. Navigation pointer for \ref{situation-setup} used inside the counted span (line 1703); the argument that consumes it is retained in the span.

Lemma 0.1. *Let (S, δ) be as in Situation 02QL. Let X be locally of finite type over S. Let $\mathcal{F}$ be a coherent $\mathcal{O}_X$-module.*

(1) *The collection of irreducible components of the support of $\mathcal{F}$ is locally finite.*
(2) *Let $Z' \subset Supp(\mathcal{F})$ be an irreducible component and let $\xi \in Z'$ be its generic point. Then*
$$length_{\mathcal{O}_{X,\xi}} \mathcal{F}_\xi < \infty$$
(3) *If $\dim_\delta(Supp(\mathcal{F})) \leq k$ and $\xi \in Supp(\mathcal{F})$ with $\delta(\xi) = k$, then ξ is a generic point of an irreducible component of $Supp(\mathcal{F})$.*

Proof. By Cohomology of Schemes, Lemma 01Y5 the support Z of $\mathcal{F}$ is a closed subset of X. We may think of Z as a reduced closed subscheme of X (Schemes, Lemma 01J3). Hence (1) follows from Divisors, Lemma 0BE1 applied to Z and (3) follows from Lemma 02QT applied to Z.

Let $\xi \in Z'$ be as in (2). In this case for any specialization $\xi' \rightsquigarrow \xi$ in X we have $\mathcal{F}_{\xi'} = 0$. Recall that the non-maximal primes of $\mathcal{O}_{X,\xi}$ correspond to the points of X specializing to ξ (Schemes, Lemma 01J7). Hence $\mathcal{F}_\xi$ is a finite $\mathcal{O}_{X,\xi}$-module whose support is $\{\mathfrak{m}_\xi\}$. Hence it has finite length by Algebra, Lemma 00L5. □